Fiscal Year:	FY 2024	**Task Last Updated:**	FY 04/15/2024
PI Name:	Costes, Sylvain Ph.D.		
Project Title:	Characterization of Female Reproductive Health Risks for Long-Duration Spaceflight Using Federated Machine Learning		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	sylvain.costes@bmsis.org	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	650-604-5343
Organization Name:	Blue Marble Space Institute of Science		
PI Address 1:	600 1st Avenue, 1st Floor		
PI Address 2:			
PI Web Page:			
City:	Seattle	**State:**	WA
Zip Code:	98104	**Congressional District:**	7
Comments:	NOTE: After retiring from NASA, the PI joined Blue Marble. Dr. Costes was previously at Lawrence Berkeley National Laboratory until December 2016.		
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBAS E.11: Animal Studies
Start Date:	06/01/2023	**End Date:**	12/31/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2024 per F. Hernandez/ARC (Ed., 4/17/24).		
Key Personnel Changes/Previous PI:	Per the PI team, the list of Co-Investigators has changed since the grant was awarded. Drs. Hoarfrost and Mackintosh are not with the project so those names have been removed from the list (Ed., 4/25/24).		
COI Name (Institution):	Cekanaviciute, Egle Ph.D. (NASA Ames Research Center) Sanders, Lauren Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The central objectives of this proposal are to study the effects of spaceflight and space-relevant radiation exposure on female mammalian reproductive health and identify biomarkers that can be used to monitor female reproductive health during space travel. The methods proposed to accomplish this objective are ground-based irradiation of female rats and spaceflight exposure of rat oocytes, subject to biomarker collection and countermeasure testing. Additionally, the proposal includes a computational machine learning component in which space-generated data are used in conjunction with Earth-based data to train a deep learning biomarker model. The proposed work is significant to the solicitation objectives in that successful completion of this work will enhance our understanding of how animals respond to stressors in space and spaceflight-like environments, through studying the fundamental biological changes that the female mammalian reproductive system undergoes while in space or exposed to space-relevant radiation. This work also enables the identification of translational biomarkers that can be used to support human female astronaut health and the development of countermeasures for helping female astronauts thrive in Low Earth Orbit missions and deep space exploration. Further, this work is significant to NASA goals of increasing diversity and representation in the astronaut cohort by focusing on a highly understudied aspect of female health in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our project will improve our knowledge of the female reproductive system in general, with benefits for Earth including better understanding of environmental stressors on female reproductive health.
Task Progress:	Since the beginning of the project, we have collected the left and right ovary from 24 4-month-old Long-Evans female rats. For 28 days, 6 of the rats were subjected to 1G and sham radiation; 6 were subjected to 2G centrifugation; 6 were subjected to 1G centrifugation and two doses of 137Cs gamma (50cGy); 6 were subjected to 2G centrifugation and two doses of 137Cs gamma (50cGy). The left ovary was preserved in optimal cutting temperature (OCT) and stored at -80C. The right ovary was preserved in Thermo Fisher RNALater and stored at -20C. From each of the left ovaries, we have generated cryosections. From these sections, we have generated Hematoxylin and Eosin (H&E) stained images for all left ovaries. These will be used to quantify the number and size of follicles in each ovary. We have also generated images from sections stained for 53bp1, a marker of DNA damage. These will be used to quantify the number of DNA damage foci in each ovarian structure.
Bibliography Type:	Description: (Last Updated: 06/03/2026)
Articles in Peer-reviewed Journals	Vinken M, Grimm D, Baatout S, Baselet B, Beheshti A, Braun M, Carstens AC, Casaletto JA, Cools B, Costes SV, De Meulemeester P, Doruk B, Eyal S, Ferreira MJS, Miranda S, Hahn C, Akyüz SH, Herbert S, Krepkiy D, Lichterfeld Y, Liemersdorf C, Krüger M, Marchal S, Ritz J, Schmakeit T, Stenuit H, Tabury K, Trittel T, Wehland M, Zhang YS, Putt KS, Zhang ZY, Tagle DA. "Taking the 3Rs to a higher level: Replacement and reduction of animal testing in life sciences in space research." Biotechnol Adv. 2025 Jul-Aug;108574. https://doi.org/10.1016/j.biotechadv.2025.108574 ; PubMed PMID: 40180136; PubMed Central PMCID: PMC12048243 , Jul-2025